

Chapter 7

$$V_{bi} = \frac{e}{2\epsilon_s} (N_d X_n^2 + N_a X_p^2)$$

势垒电容：反偏产生额外电荷

结击穿：齐纳：重掺杂时，隧穿 负

雪崩：轻掺杂时 正温度系数 电压大

Chapter 8

n_p : p区电子 p_n : n区空穴

准费米能级的变化

Chapter 9

Chapter 10

耗尽层厚度

- 1) The dominant carrier scattering mechanism in lowly-doped (high-purity) silicon at room temperature is:
 - A. Carrier-carrier scattering
 - B. Lattice scattering
 - C. Impurity ion scattering
 - D. Lattice scattering and impurity ion scattering
- 2) At room temperature, a p-type silicon sample is uniformly doped at $N_A = 10^{15}/\text{cm}^3$. Under uniform illumination such that $\Delta n = \Delta p = 10^{14}/\text{cm}^3$, calculate the resistivity if $\mu_n = 1300 \text{ cm}^2/\text{Vs}$, $\mu_p = 500 \text{ cm}^2/\text{Vs}$, $n_i = 10^{10}/\text{cm}^3$, and $q = 1.6 \times 10^{-19} \text{ C}$:
 - A. $9.191 \Omega \text{ cm}$
 - B. $9.921 \Omega \text{ cm}$
 - C. $12.50 \Omega \text{ cm}$
 - D. $48.08 \Omega \text{ cm}$
- 3) In the neutral regions near the depletion region of a forward biased pn junction, the dominant "carrier actions" are:
 - (A) Drift and diffusion.
 - (B) Generation and recombination.
 - (C) Generation and diffusion.
 - (D) Recombination and diffusion.
 - (E) Generation and drift.
 - (F) Recombination and drift.

(1) B (2) A (3) A

- 4) In a metal-semiconductor Schottky diode, current flow under forward bias consists primarily of:
- Majority carriers injected from M- $\rightarrow$ S.
 - Minority carriers injected from S- $\rightarrow$ M.
 - Majority carriers injected from S- $\rightarrow$ M.
 - Minority carriers injected from M- $\rightarrow$ S.
- 5) The threshold voltage of a MOS capacitor is **NOT** affected by:
- Doping concentration of the semiconductor.
 - Oxide charges Q_{ss}

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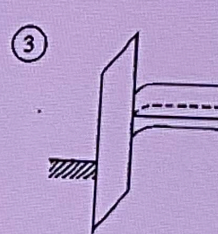
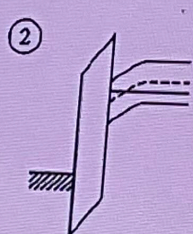
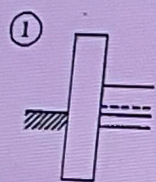
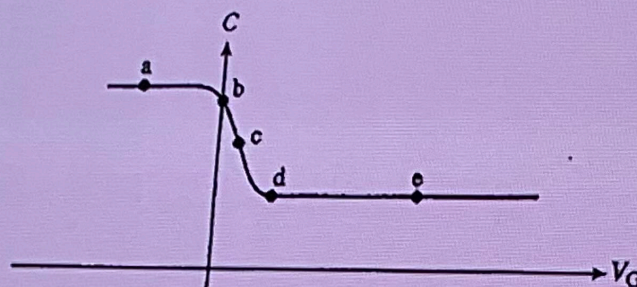
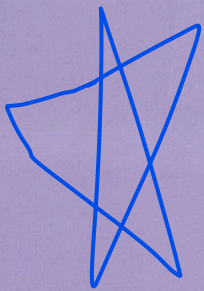
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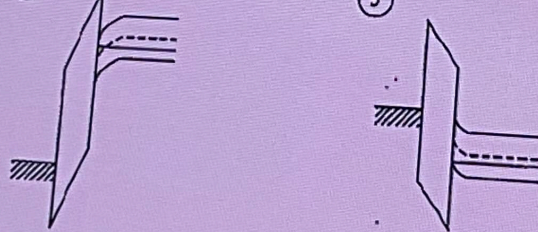
- Mobility of the carriers.
- Work function of the metal gate.
- Thickness of the oxide.
- Dielectric constant of the oxide.

(4) C (15)

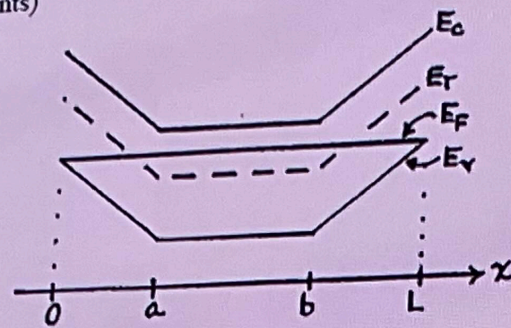
6. For an ideal MOS capacitor, (1) complete the following table making use of the ideal-structure C-V characteristics and energy band diagrams in the figures below ($1 \times 10 = 10$ points), and (2) draw the MOS capacitor block charge diagram for each case ($2 \times 5 = 10$ points).

Bias Condition	Capacitance (a-e)	Band Diagram (1-5)
Inversion	e	(4)
Depletion	c	(3)
Flat band	b	(2)
$V_G = V_T$	d	(1)
Accumulation	a	(5)

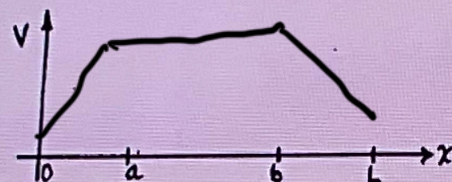




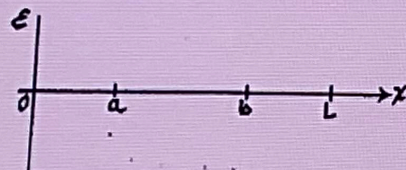
2. A Si sample maintained at room temperature is characterized by the energy band diagram of below figure. (16 points)



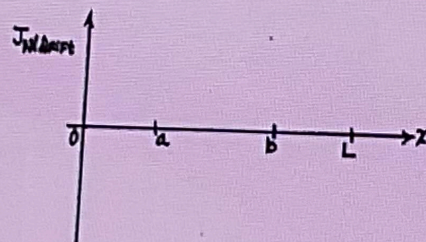
- 1) Is the semiconductor degenerate at any point? Is so, where? (2 points) 没有
- 2) The semiconductor is in equilibrium. How does one deduce this fact from the given energy band diagram? (2 points) 费米能级平坦
- 3) Sketch the electrostatic potential (V) inside the semiconductor as a function of x . (4 points)



- 4) Sketch the electric field (E) inside the semiconductor as a function of x . (4 points)

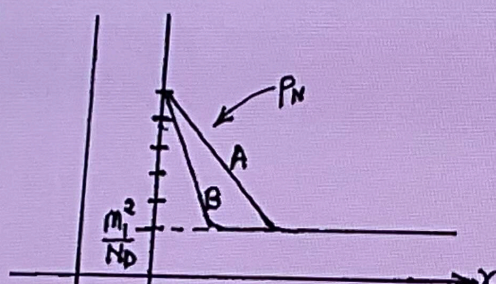


- 5) Crudely sketch the electron drift current density inside the semiconductor as a function of x . (also explain how you arrived at your sketch.) (4 points)



3. A semilog plot of the minority carrier concentration on the n-side of two ideal p^+-n diodes (A and B) maintained at room temperature is pictured below. The n-side doping (N_D) and the cross-section area are the same in both diodes. Assume low-level injection conditions prevail. (16 points)

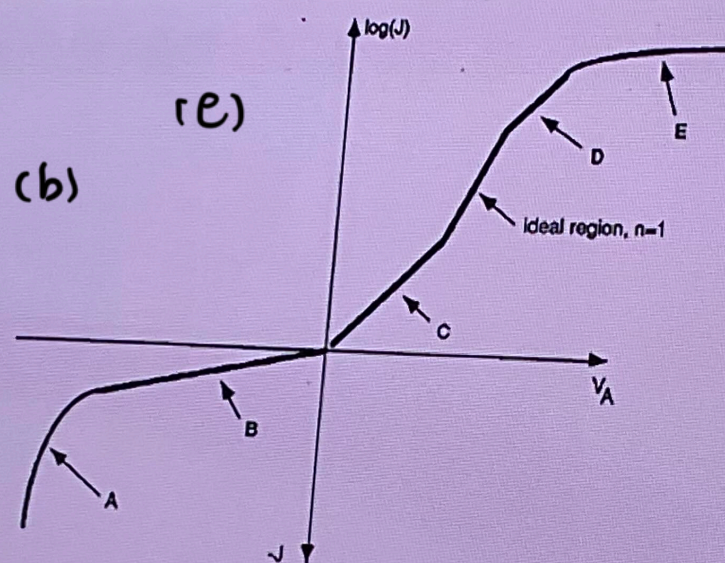
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- 1) The diodes are (i) forward biased, (ii) zero biased, or (iii) reverse biased? (4 points) (i)
- 2) The magnitude of the bias applied to diode A is (i) larger than, (ii) the same as, or (iii) smaller than the magnitude of the bias applied to diode B? (4 points) (ii)
- 3) If the diodes were to exhibit reverse bias breakdown by the avalanching process, the breakdown voltage (V_{BR}) of diode A would be (i) larger than, (ii) roughly the same as, (iii) less than the breakdown voltage of diode B? Explain how you arrive at your answer. (4 points) (iii)
- 4) The magnitude of the current flowing through diode A is (i) larger than, (ii) roughly the same as, or (iii) smaller than that of diode B? (4 points) (iii)

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4. Below is the current-voltage characteristic of a real pn junction diode. Match the possible source of the non-idealities, a-f, to the proper label, A-F) a) low injection, b) tunneling and/or avalanche current, c) recombination in depletion region, d) generation in depletion region, e) series resistance, f) high injection. (12 points)



5. Answer the following questions assuming the Al-Si contact is ideal and $N_D = 8 \times 10^{16} / \text{cm}^3$ is the doping concentration of Si. $T = 300\text{K}$, $\Phi_M(\text{Al}) = 4.28 \text{ eV}$, $\chi(\text{Si}) = 4.03 \text{ eV}$, $E_G(\text{Si}) = 1.12 \text{ eV}$, $n_i = 10^{10} / \text{cm}^3$. (16 points)
- 1) Determine Φ_S , the Si work function. (2 points)
 - 2) Determine Φ_B , the Schottky barrier diode. (2 points)
 - 3) Determine V_{bi} , the built-in potential barrier. (4 points)
 - 4) Roughly to scale, sketch the energy band diagram for the Al-Si contact under equilibrium conditions. Label relevant energy levels. (4 points)
 - 5) The Al-Si contact described in this problem is not really optimum for contacting the

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fabricated Si devices. Indicate what must be done to achieve a low-resistance Al-Si contact. (4 points)

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